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College of Engineering & Architecture
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EECE 306-01: Fundamentals of Electromagnetics Laboratory
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Lab 4 Companion: Symbolic Verification Reference

This sheet is optional and nothing in the graded work requires it. When your quadrature engine and a formula disagree, a symbolic tool is a third, independent opinion.

1. The two closed forms behind this lab

Both aspect ratio studies in Lab 4 converge toward limits that can be checked exactly.

Geometry	Field at the observation point	Limit
Finite line, length L , midplane, distance d	$E_\rho = \frac{\rho_L}{4\pi\epsilon_0 d} \frac{2L}{\sqrt{L^2 + 4d^2}}$	$\frac{\rho_L}{2\pi\epsilon_0 d}$ as $L/d \rightarrow \infty$
Disk, radius a , on axis, height h	$E_z = \frac{\rho_S}{2\epsilon_0} \left(1 - \frac{h}{\sqrt{h^2 + a^2}}\right)$	$\frac{\rho_S}{2\epsilon_0}$ as $a/h \rightarrow \infty$

Note how differently the two approach their limits, the disk deficit falls only as h/a while the line deficit falls as $(d/L)^2$. Your two ratio plots in this lab make exactly that comparison.

2. Three ways to check an integral symbolically

1. **Wolfram Alpha**, in any browser, no installation, typed in plain text.

```
integrate 1/(d^2+z^2)^(3/2) dz from -L/2 to L/2
integrate r/(r^2+h^2)^(3/2) dr from 0 to a
```

2. **Python with SymPy**, free, and the code below runs as shown.

```
import sympy as sp
z, d, L, h, a, r = sp.symbols('z d L h a r', positive=True)
I1 = sp.integrate(1/(d**2 + z**2)**sp.Rational(3,2), (z, -L/2, L/2))
print(sp.simplify(I1)) # 2*L/(d**2*sqrt(L**2 + 4*d**2))
I2 = sp.integrate(r/(r**2 + h**2)**sp.Rational(3,2), (r, 0, a))
print(sp.simplify(I2)) # 1/h - 1/sqrt(a**2 + h**2)
```

3. **The Octave symbolic package**, optional, wraps SymPy so Python must be present. Install once with `pkg install -forge symbolic`, then

```
pkg load symbolic
syms z d L positive
simplify(int(1/(d^2+z^2)^(3/2), z, -L/2, L/2))
```

What a symbolic answer can and cannot settle

A closed form is a statement about real numbers, your toolbox is a statement about floating point numbers on a finite element list. The symbolic result gives the target, the convergence study says whether your quadrature walks toward it at the advertised rate. A plausible number on a single grid proves nothing.